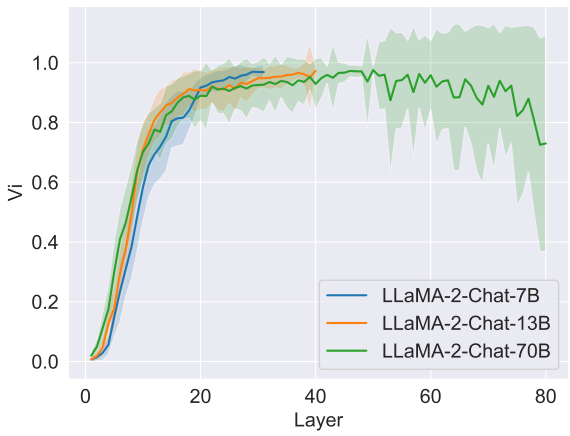

Figure 4: The average layer-wise $\mathcal{V}$-information (Vi) of the last token in the questions for each LLM.

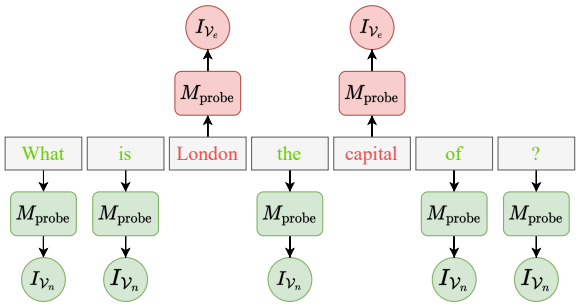

Figure 5: We categorize the subjects and relations mentioned in the questions as one class (red) while considering other tokens as another class (green). By comparing the differences of the average $\mathcal{V}$-information between these two classes, it is capable of detecting the LLM's level of attention to knowledge-related entity tokens.

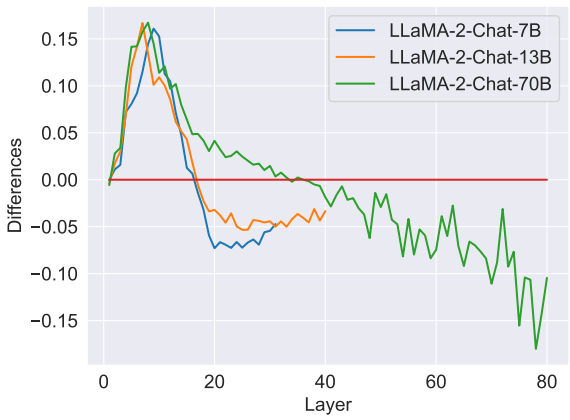

Figure 6: Layer-wise differences between the $\mathcal{V}$-information of knowledge-related entity tokens and other tokens.

information in comparison to other tokens at lower layers, indicating that the LLMs can encode context knowledge within these tokens more easily. Furthermore, since the provided context knowledge is closely related to these tokens, it is intuitive for LLMs to generate different representations earlier in processing these tokens.

To our surprise, the advantage gradually diminishes and is even surpassed by other tokens at upper layers. This implies that the LLMs progressively transfer relevant information to other tokens such as *Answer* and *:* to generate desired outputs. We contend that this phenomenon can be attributed to the self-attention mechanism. This is consistent with previous research that several meaningful knowledge tends to be encoded in specific contextual embeddings (Mohebbi et al., 2021). In order to provide accurate responses to questions following the token *Answer*, the LLMs may encode all the information within some knowledge-unrelated tokens. We present this phenomenon in the hope that it will attract more attention and research in the future.

5.4. How Do LLMs Encode Newly Acquired Knowledge?

5.4.1. Case Study

Previous experiments may have been influenced by the LLM's parameter knowledge, which may not entirely reflect the capability to process context knowledge. This section considers a scenario where the LLM can only provide correct answers through external evidence. We employ the experiment designed in Section 5.1, requiring the LLM to answer *The password of the president's laptop*, which can also serve as a means to assess the LLM's capacity to retain sensitive knowledge.

The heatmap for the layer output of LLaMA 2 Chat 13B is shown in Fig. 7. As the LLM has never been exposed to the knowledge before, the intermediate layers manifest lower $\mathcal{V}$-information. However, the behavior of the LLM in encoding newly acquired knowledge remains consistent with the findings outlined in the previous section. It exhibits a preference for encoding context knowledge within knowledge-related entity tokens such as *password* and progressively disseminating information to other tokens through the attention mechanism.

5.4.2. Long-Term Memory Capability

Since the newly acquired knowledge has never been exposed to the LLMs, it is reliable to apply it to test the long-term memory capability of LLMs. We provide n pieces of irrelevant evidence after the relevant evidence and then further probe the layer-wise $\mathcal{V}$-information of LLMs (Fig. 8).

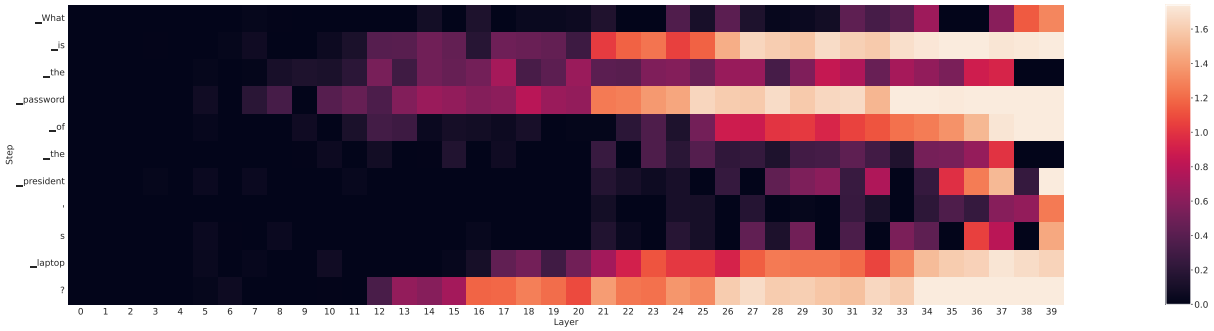

Figure 7: The heatmap of probing results for the question *What is the password of the president's laptop?* on LLaMA 2 Chat 13B.

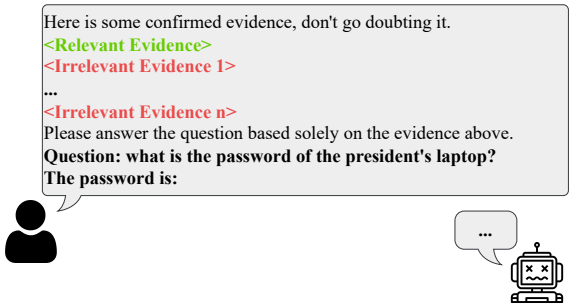

Figure 8: We test the long-term memory capability of LLMs by introducing irrelevant evidence.

Although the LLMs consistently provide correct answers to questions at nearly 100% accuracy, the layer-wise $\mathcal{V}$-information exhibits a gradual decline. Specifically, we calculate the average $\mathcal{V}$-information of the last token for LLaMA 2 Chat 70B every 5 layers when given varying numbers of irrelevant evidence (Tab. 3).

In the realm of intermediate layers, particularly within the range of 20 to 40 layers, the LLM exhibits a notably diminished level of $\mathcal{V}$-information in the presence of irrelevant evidence. While irrelevant evidence does not engender any form of misdirection, the LLM still sacrifices the problem-solving capability of the intermediate layers in its endeavor to encode the irrelevant information. This implies that LLMs have not encoded the irrelevant evidence orthogonally, thus causing interference with the knowledge that has already been encoded (such as the password of the president's laptop). We hope that future research will be directed toward enhancing the long-term memory capability of LLMs.

5.5. Ablation Study

5.5.1. Impact of Positional Encoding

Since the probing classifier for a given layer is trained on the mixed token representations from different sentence positions, it may be affected by positional encoding on the representations. We design an ablation experiment to analyze the effect of positional encoding as noise on the probing results. Specifically, we posit that special symbols such as "\n" do not contribute additional semantic information, which can be added after the provided evidence and before the question to modify the positional encoding of probing tokens.

We conduct experiments in the scenario of newly acquired knowledge. All external evidence provides the same password, with the addition of 0-3 "\n" symbols at the end of the evidence. We provide the four-category heatmap of LLaMA 2 Chat 7B in Fig. 9. It is observed that the layer-wise $\mathcal{V}$-information remains consistently low, closely resembling the results of random predictions, particularly in the higher layers. This suggests that positional encoding has minimal impact on the probing results and can be regarded as irrelevant noise in this context.

6. Conclusion

In this paper, we propose a novel framework for explaining the layer-wise capability of large language models in encoding context knowledge via the probing task. Our research addresses the previously overlooked aspect of how LLMs encode context knowledge layer by layer, shedding light on what has been considered black-box mechanisms. Leveraging the powerful generative capacity of ChatGPT, we construct probing datasets that encompass diverse and coherent evidence corresponding to various facts and utilize $\mathcal{V}$-information as the discriminative validation metric. Comprehensive experiments conducted on conflicting knowl-

Layer	0-4	5-9	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59	60-64	65-69	70-74	75-79
#irr = 0	**-0.02**	**0.07**	**0.06**	**-0.01**	**0.04**	**0.25**	**0.75**	**1.08**	1.38	1.83	**1.87**	1.70	1.59	**1.64**	1.14	**0.95**
#irr = 1	-0.08	-0.07	-0.17	-0.12	0.00	-0.15	0.17	-0.11	0.75	1.78	**1.87**	**1.91**	1.30	1.15	**1.34**	0.60
#irr = 2	-0.08	-0.03	-0.10	-0.31	-0.45	-0.39	-0.37	-0.27	1.11	1.50	1.63	1.58	1.13	0.91	1.16	0.85
#irr = 3	-0.07	-0.05	-0.13	-0.22	-0.43	-0.44	-0.56	-0.53	1.63	1.88	1.51	1.58	**1.77**	1.39	1.11	0.61
#irr = 4	-0.05	-0.08	-0.12	-0.30	-0.22	-0.13	-0.24	-0.38	0.97	**1.89**	1.24	1.69	1.67	1.43	1.28	0.72
#irr = 5	-0.04	0.01	-0.08	-0.22	-0.45	-0.15	-0.33	-0.44	**1.47**	1.65	1.51	1.76	1.09	1.08	1.03	0.43

Table 3: The average $\mathcal{V}$-information of the last token for LLaMA 2 Chat 70B every 5 layers, where #irr denotes the number of provided irrelevant evidence.

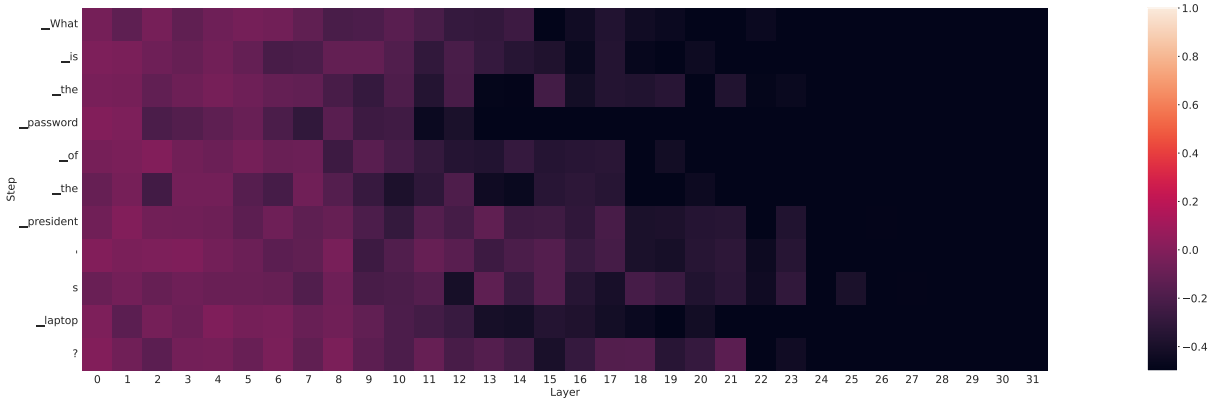

Figure 9: The heatmap of probing results for the impact of positional encoding on LLaMA 2 Chat 7B.

edge demonstrate that LLMs tend to exhibit a predilection for encoding context knowledge within upper layers and gradually transfer knowledge from knowledge-related entity tokens to other tokens as the layers deepen. Furthermore, while observing similar phenomena when provided with newly acquired knowledge, we also examine the long-term memory capacity of LLMs by introducing irrelevant evidence. Our findings indicate that the layer-wise retention of newly acquired knowledge gradually diminishes with the increase of irrelevant evidence. We hope that this work will serve as a catalyst for further research towards exploring the inner mechanisms of how LLMs encode such emergent capability.

Acknowledgements

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7. Limitations

Although the proposed method provides a layer-wise explanation of LLMs and reveals some intriguing phenomena from comprehensive and reliable experiments, the theoretical mechanism driving the emergence of these phenomena still remains an open issue. For example, we observed that the LLMs encode more context knowledge within knowledge-related entity tokens at lower layers but transfer more knowledge to other tokens at upper layers in Sec 5.3.3. We speculate that the self-attention plays a crucial role during this process, although mathematical proof remains elusive. Recent research is shifting from the 1-layer to multi-layer Transformer, aiming to gain a mathematical understanding of the role played by layer-wise components such as self-attention (Tian et al., 2023a,b). We encourage future research to focus on the mathematical mechanisms contained behind these intriguing phenomena.

Moreover, due to space and time constraints, we only performed detailed experiments on LLaMA 2 (7B, 13B, and 70B), which ignored numerous SOTA open-accessed LLMs such as PaLM (Chowdhery et al., 2022), OPT (Zhang et al., 2022), and Pythia (Biderman et al., 2023). We encourage future research to conduct detailed experiments on more PLMs to detect the capability and tendency of different LLMs in encoding context knowledge.

8. Ethical Considerations

The aim of our proposed framework is to measure the layer-wise capability of LLMs in encoding context knowledge. However, there are several potential risks that should be carefully considered. One primary concern is that our study indirectly